



MATHS SPECIAL BATCH

Algebra Sheet-4

GaganPratap Sir



- If $x^2 - \sqrt{8}x + 1 = 0$, then the value of $(x^4 - \frac{1}{x^4})$ is:

यदि $x^2 - \sqrt{8}x + 1 = 0$ है, तो $(x^4 - \frac{1}{x^4})$ का मान क्या होगा?

(a) $24\sqrt{6}$ (b) $24\sqrt{2}$
(c) $20\sqrt{2}$ (d) $20\sqrt{6}$
- If $(x - \frac{1}{x}) = \sqrt{6}$ & $x > 1$, then what is the value of $(x^8 - \frac{1}{x^8})$?

यदि $(x - \frac{1}{x}) = \sqrt{6}$ & $x > 1$, तो $(x^8 - \frac{1}{x^8})$ का मान क्या है?

[A] $1024\sqrt{15}$ [B] $992\sqrt{15}$
[C] $998\sqrt{15}$ [D] $1012\sqrt{15}$
- If $x^2 - 2\sqrt{5}x + 1 = 0$, then what is the value of $x^5 + \frac{1}{x^5}$?

यदि $x^2 - 2\sqrt{5}x + 1 = 0$ है तो $x^5 + \frac{1}{x^5}$ का मान क्या है ?

a) $408\sqrt{5}$ b) $610\sqrt{5}$
c) $406\sqrt{5}$ d) $612\sqrt{5}$
- If $X = 4 - \frac{1}{4 - \frac{1}{4 - \frac{1}{4 - \dots \infty}}}$, then find $X^5 + \frac{1}{X^5} = ?$

(a) 656 (b) 548
(c) 732 (d) 724
- If $x - \frac{1}{x} = -6$, what will be the value of $x^5 - \frac{1}{x^5}$?

यदि $x - \frac{1}{x} = -6$ है तो $x^5 - \frac{1}{x^5}$ का मान क्या होगा?

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(a) -8898 (b) -8896
(c) -8886 (d) -8892
- If $x^2 + 13x + 39 = 0$, then $(x + 8)^5 - \frac{1}{(x+8)^5} = ?$

a) 393 b) 396
c) 392 d) 394
- $x^2 - 4x + 1 = 0$, then what is the value of $(x^6 + x^{-6})$?

(a) 2786 (b) 2702
(c) 2716 (d) 2744
- If $x + \frac{1}{x} = 7$, then the value of $x^6 + \frac{1}{x^6}$ is?

यदि $x + \frac{1}{x} = 7$, तो $x^6 + \frac{1}{x^6}$ का मान क्या है?

[A] 113682 [B] 103682
[C] 103882 [D] 103862
- If $(a + \frac{1}{a}) = 7\sqrt{3}$, then what is the value of $(a^6 + a^{-6})$?

यदि $(a + \frac{1}{a}) = 7\sqrt{3}$ है, तो $a^6 + a^{-6}$ (का मान ज्ञात कीजिए)

(a) 3048190 (b) 3048542
(c) 3048132 (d) 3048625
- If $(x - \frac{1}{x} = 4)$, then what is the value of $(x^6 + \frac{1}{x^6})$?

यदि $(x - \frac{1}{x} = 4)$, तो $(x^6 + \frac{1}{x^6})$ का मान क्या है?

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- (a) 4689 (b) 4786
(c) 5832 (d) 5778
11. If $\left(x + \frac{1}{x}\right) = 5\sqrt{2}$ & $x > 1$, what is the value of $\left(x^6 - \frac{1}{x^6}\right)$?
यदि $\left(x + \frac{1}{x}\right) = 5\sqrt{2}$ & $x > 1$, तो $\left(x^6 - \frac{1}{x^6}\right)$ का मान क्या है?
[A] $22970\sqrt{23}$ [B] $23030\sqrt{23}$
[C] $23060\sqrt{23}$ [D] $22960\sqrt{23}$
12. If $x > 0$ & $x^4 + \frac{1}{x^4} = 2207$, then find $x^7 + \frac{1}{x^7}$?
यदि $x > 0$ & $x^4 + \frac{1}{x^4} = 2207$, तो $x^7 + \frac{1}{x^7}$ का मान क्या है?
[A] 710649 [B] 710647
[C] 710654 [D] 710661
13. If $x^5 - \frac{1}{x^5} = \sqrt{7}$, then find $x^{35} - \frac{1}{x^{35}} = ?$
(a) $791\sqrt{7}$ (b) $789\sqrt{7}$
(c) $790\sqrt{7}$ (d) $792\sqrt{7}$
14. If $x + \frac{1}{x} = 4$, then, $x^7 - \frac{1}{x^7} = ?$
a) $5844\sqrt{3}$ b) $5842\sqrt{3}$
c) $5824\sqrt{3}$ d) $5822\sqrt{3}$
15. If $\left(x + \frac{1}{x} + 1\right)\left(x + \frac{1}{x} - 1\right) = 2$, then
(i) $x^{38} + \frac{1}{x^{38}} = ?$
(ii) $x^{45} + \frac{1}{x^{69}} = ?$
(iii) $x^{53} + \frac{1}{x^{53}} = ?$
(iv) $x^{75} + \frac{1}{x^{75}} = ?$
(v) $x^{84} + x^{66} + x^{30} + x^{18} + x^{12} + 3 = ?$
(vi) $x^{320} + x^{314} + x^{250} + x^{244} + x^{94} + x^{88} + x^{45} + x^{39} + x^6 + 5 = ?$
(vii) $x^{79} + x^{55} + x^{41} + x^{25} + x^{19} + x^{12} + x^5 + 2 = ?$
16. If $x + \frac{1}{x} \neq 0$ and $x^3 + \frac{1}{x^3} = 0$ then the value of $\left(x + \frac{1}{x}\right)^4$ is
a) 9 b) 12
c) 15 d) 16
17. If $x^2 + x(6 - \sqrt{3}) + 10 - 3\sqrt{3} = 0$; then find the value of $(x + 3)^{17} + \frac{1}{(x+3)^{17}} = ?$
(a) $\sqrt{3}$ (b) $-\sqrt{3}$
(c) 1 (d) -1
18. $x^2 - 3x + 1 = 0$, then find the value of $x^9 + x^7 + x^{-9} + x^{-7}$?
 $x^2 - 3x + 1 = 0$, तब $x^9 + x^7 + x^{-9} + x^{-7}$ का मान बताओ
a) 6621 b) 4414
c) 9208 d) 6069
19. If $x^2 - 4x + 1 = 0$, then what is the value of $x^9 + x^7 - 194x^5 - 194x^3$?
[A] 4 [B] -4
[C] 1 [D] -1
20. If $x^2 - 3x + 1 = 0$, then what is the value of $x^{12} + x^8 - 123x^7 - 123x^3$?
(a) 7 (b) -3



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- (c) -7 (d) 3
21. If $x^3 - \frac{1}{x^3} = \sqrt{k^2 - 4}(k^2 - 1)$, then $x^2 - \frac{1}{x^2} = ?$
(a) $k\sqrt{k^2 - 4}$ (b) $k\sqrt{k^2 + 4}$
(c) $k(k^2 - 4)$ (d) $k(k^2 + 4)$
22. If $x\sqrt{x} + 3\sqrt{x} + \frac{1}{a^3} - a^3 = 0$, then find $(a^2 + \frac{1}{a^2} - x) = ?$
(a) 3 (b) 2
(c) 1 (d) 4
23. If $x - \frac{1}{x} = 1$, then $\left\{ \frac{1}{x-1} - \frac{1}{x+1} + \frac{1}{x^2+1} - \frac{1}{x^2-1} \right\} = ?$
(a) $\pm\sqrt{5}$ (b) $\frac{2}{5}$
(c) $\pm\frac{2}{\sqrt{5}}$ (d) $\pm\frac{\sqrt{5}}{2}$
24. If $x\sqrt{x} + \frac{1}{x\sqrt{x}} = 110$, then find $x^2\sqrt{x} + \frac{1}{x^2\sqrt{x}} = ?$
यदि $x\sqrt{x} + \frac{1}{x\sqrt{x}} = 110$ है, तो $x^2\sqrt{x} + \frac{1}{x^2\sqrt{x}}$ ज्ञात कीजिये?
(a) 2525 (b) 12098
(c) 140 (d) 3645
25. If $x + \frac{1}{x} = 4, x > 0$, then find the value of $x^{10} + \frac{1}{x^{10}}$.
 $x + \frac{1}{x} = 4, x > 0$, तो $x^{10} + \frac{1}{x^{10}}$ का मान ज्ञात कीजिए।
(a) 524176 (b) 524174
(c) 524172 (d) 524178
26. If $K + \frac{1}{K} - 2 = 0$ and $K > 0$, then what is the value of $K^{17} + \frac{1}{K^{12}}$?
यदि $K + \frac{1}{K} - 2 = 0$ और $K > 0$, तो $K^{17} + \frac{1}{K^{12}}$ का मान क्या है?
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(a) 2 (b) 1
(c) 0 (d) 12
27. If $x + \frac{1}{x} = 2$, then the value of $x^{11} + \frac{1}{x^{20}}$?
यदि $x + \frac{1}{x} = 2$, तो $x^{11} + \frac{1}{x^{20}}$ का मान क्या है?
[a] 7 [b] 2
[c] 1 [d] 0
28. If $x + \frac{1}{x-9} = 11$, then $(x-11)^{12} + \frac{1}{(x-11)^{13}} = ?$
(a) 0 (b) 2
(c) -2 (d) 1
29. If $a + \frac{1}{a-4} = 6$, then $(a-3)^7 + \frac{1}{(a-7)^3} = ?$
(a) $63\frac{7}{8}$ (b) $255\frac{1}{8}$
(c) $127\frac{7}{8}$ (d) 216
30. If $x + \frac{1}{x} = -2$, then $x^{2n+1} + \frac{1}{x^{2n+4}} = ?$ (where n is an integer)
(a) 2 (b) 4
(c) -2 (d) 0
31. If $K + \frac{1}{K} + 2 = 0$ and $K < 0$, then what is the value of $K^{17} + \frac{1}{K^{11}}$?



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यदि $K + \frac{1}{K} + 2 = 0$ और $K < 0$, तो $K^{17} + \frac{1}{K^{11}}$ का मान क्या है?

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- (a) -17 (b) -2
(c) -1 (d) 0

32. If $x + \frac{1}{x+5} = -7$ then $(x+5)^{19} - \frac{1}{(x+7)^{28}} = ?$

- a) 0 b) 2
c) -2 d) 1

33. If $3x + \frac{1}{3x} + 2 = 0$, then $243x^7 + \frac{1}{81x^2} = ?$

- (a) 2 (b) 4
(c) 0 (d) 1

34. If $\frac{x}{5} + \frac{5}{x} = -2$, then what is the value of x^3 ?

- a) -125 b) -25
c) $\frac{1}{125}$ d) 625

35. If $x + \frac{1}{x} = 1$,

(i) $x^{78} + \frac{1}{x^{78}} = ?$

(ii) $x^{55} + \frac{1}{x^{55}} = ?$

(iii) $x^{50} + \frac{1}{x^{50}} = ?$

(iv) $x^{71} + \frac{1}{x^{71}} = ?$

(v) $x^{34} + \frac{1}{x^{34}} = ?$

(vi) $x^{203} + x^{200} + x^{188} + x^{185} + x^{140} + x^{137} + x^{102} + x^{99} + x^{100} + x^{98} = ?$

36. If $\frac{x}{y} + \frac{y}{x} = 1$, and $x, y \neq 0$, then find the value of $x^6 + y^6 + 2x^3y^3$.

यदि $\frac{x}{y} + \frac{y}{x} = 1$, और $x, y \neq 0$ हैं, तो $x^6 + y^6 + 2x^3y^3$ का मान ज्ञात करें।

- (a) 0 (b) 1
(c) x^3y^3 (d) $3x^3y^3$

37. If, $\frac{r}{13} + \frac{13}{r} = 1$ then the value of r^3 is:

यदि, $\frac{r}{13} + \frac{13}{r} = 1$ है, तो r^3 का मान ज्ञात कीजिए।

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- (a) -2157 (b) 2197
(c) 2157 (d) -2197

38. If $x + \frac{1}{x} = 1$, then $(1 + x + x^2)(1 - x - x^2) = ?$

- (a) 0 (b) 4
(c) -2 (d) 1

39. If $4x^2 + \frac{1}{x^2} = 2$, then find $8x^3 + \frac{1}{x^3} = ?$

- (a) 0 (b) 1
(c) 2 (d) 4

40. If $x + \frac{1}{x} = 1$, then the value of $x^{12} + x^9 + x^6 + x^3 + 1$ is:

यदि $x + \frac{1}{x} = 1$ है, तो $x^{12} + x^9 + x^6 + x^3 + 1$ का मान क्या है?

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- (a) 1 (b) -1



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- (c) 0 (d) -2
41. If $x^5 + \frac{1}{x^5} = 1$, then $x^{180} + x^{165} + x^{145} + x^{130} - 2x^{105} + x^{80} + x^{70} - x^{15} + 3 = ?$
 (a) 3 (b) 6
 (c) 4 (d) 5
42. If $(x+2)^2 + \frac{1}{(x+2)(x+3)-(x+1)} = 0$, then find $(x^2 + 4x + 5)^3$?
 यदि $(x+2)^2 + \frac{1}{(x+2)(x+3)-(x+1)} = 0$ है, तो $(x^2 + 4x + 5)^3$ ज्ञात कीजिए?
 [a] 0 [b] 1
 [c] -1 [d] 343
43. If $x + \frac{3}{x} = 3$, then find $(x-1)^8 + \frac{1}{(x-1)^8}$?
 यदि $x + \frac{3}{x} = 3$ है, तो $(x-1)^8 + \frac{1}{(x-1)^8}$ ज्ञात कीजिये ?
 a) 1 b) -1
 c) 2 d) 0
44. If $\frac{x}{5\sqrt{6}} + 1 = \frac{-5\sqrt{6}}{x}$, then find the value of $x^5 - 750\sqrt{6}x^2 + 1 = ?$
 यदि $\frac{x}{5\sqrt{6}} + 1 = \frac{-5\sqrt{6}}{x}$, तो $x^5 - 750\sqrt{6}x^2 + 1$ का मान ज्ञात कीजिए?
 A) $25\sqrt{6}$ B) 1
 C) 0 D) $1+5\sqrt{6}$
45. If $x^2 - 7x + 13 = 0$ then $(x-3)^{22} + \frac{1}{(x-3)^{22}} = ?$
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 a) 3 b) 1
 c) -1 d) -2
46. If $\frac{a}{b} + \frac{b}{a} = -1$ and $a - b = 2$, then the value of $a^3 - b^3$ is:
 यदि $\frac{a}{b} + \frac{b}{a} = -1$ और $a - b = 2$ है, तो $a^3 - b^3$ का मान ज्ञात करें।
(SSC CGL 2022)
 (a) 0 (b) $\frac{1}{2}$
 (c) 1 (d) -1
47. If $3x + \frac{1}{2x} = \sqrt{3}$, then find $6x^2 + \frac{1}{6x^2}$?
 यदि $3x + \frac{1}{2x} = \sqrt{3}$ है, तो $6x^2 + \frac{1}{6x^2}$ ज्ञात कीजिये?
 (a) -1 (b) 0
 (c) 2 (d) 1
48. If $5x + \frac{1}{2x} = \sqrt{5}$, then find $50x^4 + \frac{1}{100x^4}$?
 यदि $5x + \frac{1}{2x} = \sqrt{5}$ है, तो $50x^4 + \frac{1}{100x^4}$ ज्ञात कीजिये?
 (a) -1 (b) -1/2
 (c) -4/3 (d) -3/2
49. If $x^2 + y^2 = x^{-2}(777 - y^4)$ and $x + y = x^{-1}(37 - y^2)$, then find the value of $x^2 + y(y-x) = ?$
 यदि $x^2 + y^2 = x^{-2}(777 - y^4)$ और $x + y = x^{-1}(37 - y^2)$ तो $x^2 + y(y-x)$ का मान कीजिए?
 (a) 29 (b) 21



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(c) 23

(d) 31

50. If $x^2 - xy + y^2 = 13$ and $x^2 + xy + y^2 = 37$, then the value of $\frac{x^6 - y^6}{x^2 - y^2}$ is ?

यदि $x^2 - xy + y^2 = 13$ और $x^2 + xy + y^2 = 37$, तो $\frac{x^6 - y^6}{x^2 - y^2}$ का मान ज्ञात करें ?

a) 481

b) 500

c) 520

d) 444

51. If $x^4 + x^2y^2 + y^4 = 273$ and $x^2 - xy + y^2 = 13$, then the value of xy is?

यदि $x^4 + x^2y^2 + y^4 = 273$ और $x^2 - xy + y^2 = 13$ तो xy का मान है?

a) 4

b) 10

c) 6

d) 8

52. If $x^4 + y^4 + x^2y^2 = 117$ & $x^2 + y^2 - xy = 3(4 + \sqrt{3})$ then find $x^2 + y^2$?

यदि $x^4 + y^4 + x^2y^2 = 117$ & $x^2 + y^2 - xy = 3(4 + \sqrt{3})$ है, तो $x^2 + y^2$ ज्ञात कीजिए?

(a) 9

(b) $6\sqrt{3}$

(c) 12

(d) $13\sqrt{3}$

53. If $x^2 - xy + y^2 = 17$ and $x^4 + x^2y^2 + y^4 = 425$, then the value of $\frac{x}{y} + \frac{y}{x}$ is

यदि $x^2 - xy + y^2 = 17$ और $x^4 + x^2y^2 + y^4 = 425$ तो $\frac{x}{y} + \frac{y}{x}$ का मान क्या है?

(a) 6.25

(b) 5.25

(c) 6.4

(d) 5.5

54. If $x^2 - xy + y^2 = 7$ and $x^4 + x^2y^2 + y^4 = 105$, then the value of $\frac{x^3}{y^3} + \frac{y^3}{x^3}$ is?

यदि $x^2 - xy + y^2 = 7$ और $x^4 + x^2y^2 + y^4 = 105$ तो $\frac{x^3}{y^3} + \frac{y^3}{x^3}$ का मान क्या है?

(a) $\frac{437}{32}$

(b) $\frac{317}{16}$

(c) $\frac{713}{64}$

(d) $\frac{803}{64}$

55. If $x^4 + x^2y^2 + y^4 = \frac{21}{256}$ and $x^2 + xy + y^2 = \frac{3}{16}$, then $(x + y) = ?$

यदि $x^4 + x^2y^2 + y^4 = \frac{21}{256}$ और $x^2 + xy + y^2 = \frac{3}{16}$ है, तो $x + y = ?$

(a) $\frac{1}{16}$

(b) $\frac{5}{8}$

(c) $\frac{3}{8}$

(d) $\frac{1}{4}$

56. If $x^4 + y^4 + x^2y^2 = 17\frac{1}{16}$ and $x^2 - xy + y^2 = 5\frac{1}{4}$, then one of the values of $(x - y)$ is:

यदि $x^4 + y^4 + x^2y^2 = 17\frac{1}{16}$ और $x^2 - xy + y^2 = 5\frac{1}{4}$ तो $(x - y)$ का मान है:

(a) $\frac{5}{2}$

(b) $\frac{3}{4}$

(c) $\frac{5}{4}$

(d) $\frac{3}{2}$

57. If $1 + 9r^2 + 81r^4 = 256$ and $1 + 3r + 9r^2 = 32$, then find value of $1 - 3r + 9r^2$.

यदि $1 + 9r^2 + 81r^4 = 256$ और $1 + 3r + 9r^2 = 32$, है तो $1 - 3r + 9r^2$ का मान ज्ञात करें।

(a) 8

(b) 4

(c) 16

(d) 12



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a) $\frac{7}{12}$
c) $\frac{2}{3}$

b) 0.625
d) $\frac{5}{9}$

67. If $a^2 + b^2 = x, ab = y, \frac{a^4 + b^4}{a^2 - ab\sqrt{2} + b^2} = ?$

(a) $x + 2y$

(b) $x + \sqrt{2}y$

(c) $y + \sqrt{2}x$

(d) $2x + y$

68. If $a^4 + b^4 = 63$ & $a^2 + b^2 + ab\sqrt{2} = 9$, then find ab ?

यदि $a^4 + b^4 = 63$ & $a^2 + b^2 + ab\sqrt{2} = 9$ है, तो ab ज्ञात कीजिये?

[a] 1

[b] $1/\sqrt{2}$

[c] 2

[d] 0

69. If the semi-perimeter and area of a rectangular whose length and breadth are 'x' and 'y' is 12 cm and 28 cm², respectively, then find the value of $x^4 + x^2y^2 + y^4$.

यदि उस आयताकार मैदान की अर्ध-परिधि और क्षेत्रफल क्रमशः 12 cm और 28 cm² है, जिसकी लंबाई 'x' और चौड़ाई 'y' हो, तो $x^4 + x^2y^2 + y^4$ का मान ज्ञात कीजिए।

SSC Selection Post (Phase-XII)

[A] 6609

[B] 6906

[C] 6960

[D] 6690

70. If $x^4 + x^3 + x^2 + x + 1 = 0$, then $x^{2015} + x^{3015} + 5 = ?$

a) 6

b) 7

c) 8

d) -9

71. If $a^9 + a^8 + a^7 + a^6 + a^5 + a^4 + a^3 + a^2 + a^1 + 1 = 0$, then find $a^{2020} + \frac{1}{a^{2030}} + a^{2040} + \frac{1}{a^{2050}}?$

यदि $a^9 + a^8 + a^7 + a^6 + a^5 + a^4 + a^3 + a^2 + a^1 + 1 = 0$ है, तो $a^{2020} + \frac{1}{a^{2030}} + a^{2040} + \frac{1}{a^{2050}}$ ज्ञात कीजिये?

[a] 1

[b] 4

[c] 0

[d] 2

72. If $y = 1 + \sqrt{3} + \sqrt{4}$, then the value of $2y^4 - 8y^3 - 6y^2 + 28y - 84$?

यदि $y = 1 + \sqrt{3} + \sqrt{4}$ है, तो $2y^4 - 8y^3 - 6y^2 + 28y - 84$ का मान क्या होगा?

[a] $40\sqrt{3}$

[b] $80\sqrt{3}$

[c] $20\sqrt{3}$

[d] $60\sqrt{3}$

73. If $a^3 + 4a^2 + 16a = 1$, then what is the value of $a^3 + (\frac{4}{a})?$

[a] 63

[b] 65

[c] 67

[d] 68

74. If $a^3 + 5a^2 + 25a + 2 = 0$, then find the value of $a^3 - \frac{10}{a}?$

यदि $a^3 + 5a^2 + 25a + 2 = 0$ है, तो $a^3 - \frac{10}{a}$ का मान ज्ञात कीजिये?

[a] 127

[b] 125

[c] 25

[d] 123

75. If $x^2 + 2 = 2x$, then $x^4 - x^3 + x^2 + 2 = ?$

a) 0

b) 1

c) -1

d) $\sqrt{2}$



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76. If $3\sqrt{\frac{1-a}{a}} + 9 = 19 - 3\sqrt{\frac{a}{1-a}}$, then what is the value of a ?
- (a) $3/10, 7/10$ (b) $1/10, 9/10$
(c) $2/5, 3/5$ (d) $1/5, 4/5$
77. If $\sqrt{\frac{x}{y}} + \frac{1}{\sqrt{2}} = \sqrt{\frac{y}{x}}$ and $x + y = 18$, then what is the value of xy ?
- (a) 80 (b) 45
(c) 56 (d) 72
78. If $\sqrt{\frac{a}{b}} = \frac{8}{3} + \sqrt{\frac{b}{a}}$ & $(a + b) = 30$, then what is the value of ab ?
- यदि $\sqrt{\frac{a}{b}} = \frac{8}{3} + \sqrt{\frac{b}{a}}$ & $(a + b) = 30$, तो ab का मान क्या है?
- [a] 64 [b] 28
[c] 81 [d] 26
79. If $(x + 5)(x + 6) + \frac{9}{x(x-1)} = 0$, then $2x^2 + 10x + 7 = ?$
- (a) 10 (b) 11
(c) 9 (d) 17
80. If $x^3 - 6x^2 + 35 = 8(x - 2)$, then find the value of $x^2 + \frac{3}{(x-6)} = ?$
- यदि $x^3 - 6x^2 + 35 = 8(x-2)$, तब $x^2 + \frac{3}{(x-6)}$ का मान क्या होगा ?
- (a) 6 (b) 8
(c) 9 (d) 12
81. Given that x, y, z are positive real numbers, if $(x + y)^2 - z^2 = 130$, $(y + z)^2 - x^2 = 150$ and $(x + z)^2 - y^2 = 120$, then x is:
- दिया गया है कि x, y, z धनात्मक वास्तविक संख्याएं हैं, यदि $(x+y)^2 - z^2 = 130$, $(y+z)^2 - x^2 = 150$ और $(x+z)^2 - y^2 = 120$ है, तो x किसके समान होगा?
- (a) $\frac{35}{8}$ (b) $\frac{25}{4}$
(c) $\frac{15}{2}$ (d) $\frac{16}{3}$
82. If $x = 2 + \sqrt{3}$, $y = 2 - \sqrt{3}$, $z = 1$, then what is the value of $\left(\frac{x}{yz}\right) + \left(\frac{y}{xz}\right) + \left(\frac{z}{xy}\right) + 2\left[\left(\frac{1}{x}\right) + \left(\frac{1}{y}\right) + \left(\frac{1}{z}\right)\right]$?
- यदि $x = 2 + \sqrt{3}$, $y = 2 - \sqrt{3}$, तथा $z = 1$ है तो $\left(\frac{x}{yz}\right) + \left(\frac{y}{xz}\right) + \left(\frac{z}{xy}\right) + 2\left[\left(\frac{1}{x}\right) + \left(\frac{1}{y}\right) + \left(\frac{1}{z}\right)\right]$ का मान क्या है?
- (a) 25 (b) 22
(c) 17 (d) 43
83. If $x = (\sqrt{5}) + 1$ and $y = (\sqrt{5}) - 1$, then what is the value of $\left(\frac{x^2}{y^2}\right) + \left(\frac{y^2}{x^2}\right) + 4\left(\frac{x}{y}\right) + 4\left(\frac{y}{x}\right) + 6$?
- यदि $x = (\sqrt{5}) + 1$ तथा $y = (\sqrt{5}) - 1$ है, तो $\left(\frac{x^2}{y^2}\right) + \left(\frac{y^2}{x^2}\right) + 4\left(\frac{x}{y}\right) + 4\left(\frac{y}{x}\right) + 6$ का मान क्या है?
- (a) 31 (b) $23\sqrt{5}$



MATHS SPECIAL BATCH

Algebra Sheet-4



Gagan Pratap Sir

(c) $27\sqrt{5}$

(d) 25

84. If $\frac{ab}{a+b} = \frac{1}{3}$, $\frac{bc}{b+c} = \frac{1}{4}$, $\frac{ca}{c+a} = \frac{1}{5}$, then find the value of $\frac{abc}{ab+bc+ca}$?

यदि $\frac{ab}{a+b} = \frac{1}{3}$, $\frac{bc}{b+c} = \frac{1}{4}$, $\frac{ca}{c+a} = \frac{1}{5}$ है, तो $\frac{abc}{ab+bc+ca}$ का मान ज्ञात कीजिये?

[a] $1/6$

[b] $1/12$

[c] 6

[d] $1/4$

85. If $\frac{xy}{x+y} = a$, $\frac{xz}{x+z} = b$ and $\frac{yz}{y+z} =$

c where a, b, c are all non-zero numbers, then x equals to?

यदि $\frac{xy}{x+y} = a$, $\frac{xz}{x+z} = b$ और $\frac{yz}{y+z} = c$ जहाँ a, b, c गैर शून्यक संख्याये है तो $x = ?$

[a] $\frac{2abc}{ab+bc-ac}$

[b] $\frac{2abc}{ab+ac-bc}$

[c] $\frac{2abc}{ac+bc-ab}$

[d] $\frac{2abc}{ab+bc-ac}$

86. If $2018^x + 2018^{-x} = 3$, then find $\sqrt{\frac{2018^{6x} - 2018^{-6x}}{2018^x - 2018^{-x}}}$?

यदि $2018^x + 2018^{-x} = 3$ है, तो $\sqrt{\frac{2018^{6x} - 2018^{-6x}}{2018^x - 2018^{-x}}}$ ज्ञात कीजिये?

[a] 16

[b] 10

[c] 12

[d] 9